Related work via Gemini Notebook: setup and prompts

Oct 9, 2026 · @Minidu

What we want out of each paper

For each of the three papers we need a page-cited fact sheet, a check of the claims our documents already make about it, and a 90–120 word related-work paragraph. Together they feed the 1-page Related Work section and the positioning table (Research Approach v3.1).

Paper	Source to upload	What the paper section needs from it
NeuroAdaptX (Bunde, DESRIST 2026, LNCS 16605, pp. 205–225)	Publisher PDF	Closest competitor: what it adapts, how profiles are chosen, how equivalence is checked, N=216 result. Must be contrasted, not just cited
Tielman et al. (2024), Technology in Society 79, 102685	Journal PDF	The roadmap we answer; the quotes already used in the evidence catalogue
Madugalla et al., human-centred RE for critical systems	Both versions: arXiv:2511.12856 and SSRN 6991922 (2026)	Our domain precedent: cognitive impairment out of scope, forecast shown but not explained; settle which version and year to cite

Gemini is used to find and quote, not to judge. Every sentence that reaches the paper is checked against the page it cites (last section).

Set up the notebook

One notebook, four PDFs, and one block of custom instructions so every answer comes back cited and in the same shape. Notebooks need a personal Google account; work and school accounts are not supported (Chrome Story).

1. Go to gemini.google.com, open the sidebar, choose **New notebook**, and name it `UISE27 related work`.
2. **Add sources**: the NeuroAdaptX PDF, the Tielman 2024 PDF, and both Madugalla versions (arXiv and SSRN). Rename each source so the name says the paper and version, e.g. `Madugalla_arXiv_2511.12856` and `Madugalla_SSRN_2026`.
3. Do not upload our own documents (Research Approach, catalogue, handoff). Gemini would then quote our claims back as if the papers made them. Our context goes in the instructions instead.
4. Open the notebook, then **More → Notebook settings**, paste the block below, and **Save**. The notebook also appears in Gemini Notebook (formerly NotebookLM), synced. If you work there instead, paste the same block into its chat configuration, or as your first chat message.
5. Start a new chat for each paper, so answers about one paper do not mix into another. In that chat, tick only that paper's source (both versions for Madugalla).

```
You are helping write the related-work section of an 8-page software
engineering workshop paper. Work ONLY from the sources in this notebook.

Rules:
1. Every factual statement ends with a citation to the source and its page
number, e.g. [Tielman 2024, p. 5]. No page, no statement.
2. If a source does not state something, write "Not stated in the source". Do
not fill gaps from general knowledge, and do not guess numbers.
3. Quote exact wording only when asked, keep each quote under 25 words, and
mark it with quotation marks and a page number.
4. Keep apart: what the authors did, what they found (with numbers), what
they claim or recommend, and what they leave as future work or limitations.
Label each.
5. When a source has two versions, say which version each statement comes
from.
6. Plain English, short sentences. Use tables when asked.
7. Anything that compares a paper with OUR work is your inference. Put it
under a heading "Comparison with our work (inference)" and never mix it with
what the paper says.

Our work, for comparison only (this is NOT a source):
We adapt how a flood ML prediction is explained to people with cognitive
impairments, using a user-chosen explanation profile built from explanation
needs, not diagnosis. Requirements come from a 128-guideline accessibility
catalogue (50 judged relevant to explanations) and are traced to software
capabilities. The prediction never changes; only its explanation does. A
runtime check confirms five facts survive every adapted explanation: hazard,
place, time window, likelihood (uncertainty kept) and action. Evaluation is a
persona-based cognitive walkthrough by the authors; no user study has been
run, and no profile has been validated with users.
```

Prompts, in the order to run them

Run prompts 1 to 4 once per paper, each paper in its own chat; run 5 and 6 once, with all sources ticked. Replace `[PAPER]` with the source name. Save each answer as a note (pin it) so the notebook keeps it.

1. Reference details (run first, per paper)

```
From [PAPER] only, give the full reference: all authors in order, exact
title, venue, year, volume, issue or LNCS number, pages or article number,
DOI, arXiv or SSRN id, and the version date printed on the document. Cite the
page where you found each item. Write "Not stated in the source" for anything
missing.
```

For Madugalla, add this line:

```
There are two versions (arXiv and SSRN). Put them side by side in a table:
title, authors, date, venue, and any difference in the numbers of guidelines,
requirements, interviews, evaluators or personas. Then say which version is
newer, based only on the dates printed in them.
```

2. Fact sheet (per paper)

```
Fill this table for [PAPER]. One row per field, columns: Field | What the
paper says | Page. Keep each answer under 40 words.

Fields:
1. Problem the paper addresses
2. Domain, and whether a wrong or misunderstood output could cause harm
3. Target users, and how they are described (condition or diagnosis labels,
needs, expertise, demographics)
4. Is an AI or ML prediction being explained? Which model and which output?
5. What is adapted or recommended (content, order, structure, modality,
density, wording, interaction)
6. How the adaptation is chosen (self-report, carer, rules, a model, fixed by
the designer)
7. Where requirements or design principles come from, and how many
8. Are requirements traced to system features or capabilities? How?
9. How the paper checks that adapted versions keep the same information: what
is checked, by whom, and when (design time or run time)
10. How uncertainty or probability is shown or discussed
11. Evaluation: method, number and type of participants, task, measures
12. Main results, with the numbers
13. Limitations the authors state
14. Future work the authors state

After the table, add "Comparison with our work (inference)": at most 3
bullets on what this paper leaves open that our work addresses.
```

3. Check the claims we already make (per paper)

Paste the block for that paper. These are the claims in the Research Approach v3.1, the evidence catalogue and the handoff.

```
Check each claim below against [PAPER]. Answer in a table: Claim | Verdict
(Confirmed / Partly / Not found / Contradicted) | Exact wording from the
source, under 25 words | Page. Do not soften a verdict.

NeuroAdaptX claims:
1. Adapts explanation structure, modality and density.
2. Profiles are self-reported condition profiles (ADHD, autism, dyslexia).
3. Derives 10 design requirements from cognitive theory.
4. Randomised online vignette experiment, N = 216.
5. The task is in a digital-wellbeing setting.
6. Informational equivalence is checked with a pre-deployment checklist.
7. Adapted explanations are fixed templates; uncertainty is not part of them.
8. Published in DESRIST 2026, LNCS 16605, pp. 205–225.
```

```
Tielman et al. 2024 claims:
1. "Adaptive XAI is the key to inclusive XAI" (exact wording?).
2. Calls for personalisation that is "more than just simplicity in text"
(exact wording?).
3. People with cognitive disabilities "should be involved throughout the
development process of XAI" (exact wording?).
4. Counterfactual explanations do not necessarily lead to feasible and
actionable explanations.
5. Poses three questions: how explainable, what understanding, how to
implement.
6. Reports a lack of real-world XAI studies for this group.
7. Most existing research presents feature importance.
8. Personalisation is recognised as important but proven examples hardly
exist.
9. Does not address disasters or a requirements specification.
```

```
Madugalla et al. claims (answer for each version separately):
1. Covers four groups: older adults, low-digital-literacy users, rural users,
colour-blind users.
2. 62 guidelines turned into 67 requirements.
3. Validation by 6 interviews and cognitive walkthroughs with 8 evaluators
and 4 personas.
4. States that cognitive impairments were outside the scope of the study
(exact wording?).
5. The prototype shows a four-day fire danger forecast.
6. The forecast is shown but never explained; there is no XAI.
7. Concludes that adaptivity must always remain explainable and controllable
(exact wording?).
```

4. Related-work paragraph (per paper)

```
Using only your fact sheet and claim check for [PAPER] in this chat, write one paragraph of 90 to 120 words for a related-work section. Sentence 1: what the paper does. Sentences 2–3: how, with its key number. Last 1–2 sentences: what it leaves open, starting "It does not" and stating only gaps the paper itself shows (e.g. a population it excludes). Add the page for each sentence in brackets, which I will remove later. No praise words.
```

5. Positioning table (once, all sources ticked)

```
Make a table with one column per paper (NeuroAdaptX, Tielman 2024, Madugalla) and these rows: domain and stakes; target population and how profiles are defined; what is adapted; where requirements come from; requirement-to-feature tracing; how information equivalence or fidelity is checked and when; whether uncertainty is kept in adapted output; evaluation method and N. Each cell under 20 words with a page number, or "Not stated in the source". Do not add a column for our work.
```

6. Look for evidence against our gap (once, all sources ticked)

```
Our gap statement is: no work turns cognitive accessibility requirements into explanation requirements for safety-critical ML predictions, specifies them as user-chosen profiles, and checks at run time that adapted explanations stay faithful to the prediction. Search all sources for anything that weakens any part of it: safety-critical or disaster use, need-based or user-chosen profiles, requirements traced to features, automatic or run-time checking of adapted text, keeping uncertainty. Quote it with the page. If nothing weakens a part, say "Nothing found" for that part.
```

Before anything goes into the paper

A Gemini answer is a lead, not a source: open the cited page for every sentence you keep.

- [ ] Click each citation in the answer and confirm the page says it. Delete any sentence whose citation does not hold.
- [ ] Any "Partly", "Not found" or "Contradicted" verdict in prompt 3: fix that claim in the Research Approach and the evidence catalogue too.
- [ ] Decide the Madugalla citation (arXiv 2025 or SSRN 2026) from prompt 1 and use one version everywhere.
- [ ] Keep "Comparison with our work (inference)" out of the paper as written; rewrite it in your own words as our position.

- [ ] Nothing in a paragraph may say or imply that our profiles improve comprehension, that words plus picture is best for people with cognitive impairments, that "Possible" means 30% in CAP, or that any profile was validated.
- [ ] Save the checked fact sheets, claim tables and paragraphs as `docs/research/related-work-notes.md` in the repo, and bring the prompt 3 and prompt 6 tables back to Claude for Framework v2 (step 3).